

Worst-Case Height Comparison of Trees

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Consider a tree T storing 100,000 entries. What is the worst-case height of T in the following cases?

- T is an AVL tree
- T is a red-black tree
- T is an $(2,4)$ tree
- T is a binary search tree

a. T is an AVL tree

The height of a AVL tree is strictly less than $2 \log_2(n+2)$.

Substituting the value of $n = 100,000$

$$2 \log_2(100000 + 2)$$

$$\log_2(2^{16}) < \log_2(100000 + 2) < \log_2(2^{17})$$

$$16 < \log_2(100000 + 2) < 17$$

The approximate value of $2 \log_2(100000 + 2)$ is 33.

b. T is a red-black tree

Substituting the value of $n = 100,000$

$$2 \log_2(100000 + 2)$$

$$\log_2(2^{16}) < \log_2(100000 + 2) < \log_2(2^{17})$$

$$16 < \log_2(100000 + 2) < 17$$

The approximate value of $2 \log_2(100000 + 2)$ is 33.

c. T is a (2,4) tree

The height of a (2,4) tree is strictly less than $\log_2(n+1)$

Substituting the value of $n = 100000$

$$\log_2(2^{16}) < \log_2(100000+2) < \log_2(2^{17})$$

$$16 < \log_2(100000+2) < 17$$

Taking the upper bound

The approximate value of $\log_2(100000+2)$ is 17.

d. The worst height of the binary tree is $O(n)$

The worst case is linear. So the approximate would be 100000.